IYPT Pinhole Sunglasses

Theoretical Handbook 2 — Chapter 13 Verification Addendum

Section 13.3: Implementation Verification of the Vignetting Geometry

Contents

A note on this file 1

13 Field of View (Chapter 13 continued) **2**

13.3 Implementation Verification 2

A note on this file

This is a **new, standalone** document, not an edit to any existing handbook source. Chapter 13 ("Field of View", §13.1–13.2, the pupil-displacement and circle-overlap formulas this section verifies) exists in `IYPT_Pinhole_Theoretical_Handbook_2.pdf`, which has no LaTeX source anywhere in this project – it cannot be edited in place. `IYPT_Pinhole_Handbook_2_source.tex` (the file with an editable source) is a *different* document (single-aperture diffraction/defocus/photometry/acuity, 19 chapters, no Chapter 13) and was left untouched, per instruction. This file is written to be inserted as Section 13.3 wherever Chapter 13's source is eventually made editable, and stands on its own in the meantime. Preamble matches `IYPT_Pinhole_Handbook_2_source.tex`'s style (margins, heading colour/format, package set) for visual consistency.

Chapter 13

Field of View (Chapter 13 continued)

13.3 Implementation Verification

The §13.1/§13.2 formulas were implemented in `array_optimization/vignetting.py` as pure ray/aperture geometry, decoupled from the wave-optics propagator, and validated against an independent Monte Carlo estimate (no shared code) – converging to 0.177 percentage points of full hole area (tolerance 0.5) after an initial, statistically ill-conditioned tolerance metric (a naive relative-area ratio, which blows up as true area $\to 0$ with no actual bias) was replaced with one normalized against full hole area throughout. A boundary defect was also found and fixed: floating-point roundoff at $\Delta = R_p + R_h$ produced a small negative area ($-0.059\,\mu\text{m}^2$); the final combination is now clamped to $[0, \infty)$, and no negative values remain.

Applied to this project's 20 Pareto-optimal array candidates, gaze-angle field-averaging over $\pm 25°$ (the validated range) does **not** change Pareto-front membership for any candidate, under *both* the uniform-placeholder and the literature-grounded `gaussian_centered` distribution (saccade-amplitude/center-bias sourced; see `GAZE_DISTRIBUTION_SENSITIVITY.md` for citations). The two distributions' field-averaged transmission estimates agree to within 0.02 percentage points – a materially stronger robustness claim than one placeholder alone.

Secondary finding, held under both distributions: of the 13/20 candidates dark at primary gaze ($N_{\text{eff,worst}}(\phi{=}0) = 0$, ghosting-safe but momentarily lightless), 11–12 show measurably *increased* N_{eff} once realistic gaze movement is integrated – rotation can bring a hole into view, not only vignette one. Reported as a real result, not smoothed into "no change."

Unresolved: $w(\phi)$ is grounded in saccade-*amplitude*/center-bias literature, not a direct measurement of gaze-*dwell-time* while wearing pinhole sunglasses – flagged for future refinement, not presented as closed.

Table 13.1: Chapter 13 implementation verification: headline numbers.

Check	**Uniform**	**Gaussian** (σ=6°)
Candidates on Pareto front, static (ϕ=0)	20/20	20/20
Candidates on Pareto front, field-averaged T	20/20	20/20
Candidates with ranking changed (static vs. field-avg.)	0/20	0/20
$T_{\text{field}}/T_{\text{static}}$ range across the 20	99.918–100.184%	99.924–100.163%
Dark-at-ϕ=0 candidates ($N_{\text{eff,worst}} = 0$)	13/20	
of those, now lit under field-averaging	11/13	12/13
MC cross-check, max error (of full hole area)	0.177 pct. points (tol. 0.5)	